

Mathematics notation exercises

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22.3.2024

1. Let f be a quadratic function with nonzero second-order coefficient, that is,

$$f(x) = ax^2 + bx + c,$$

where a , b and c are real-valued constants and $a \neq 0$.

2. In $\triangle ABC$, $a = AB = 5.0\text{cm}$, $b = AC = 6.0\text{cm}$ and $\alpha = \hat{BAC} = 54.5^\circ$. Then the area A_t of the triangle is

$$\begin{aligned} A_t &= \frac{1}{2}ab \sin \alpha \\ &= \frac{1}{2} \times 5\text{cm} \times 6\text{cm} \times \sin 54.5^\circ \\ &\approx 12\text{cm}^2. \end{aligned}$$

3. The finite geometric sum

$$S_n = a + aq + \cdots + aq^{n-1}$$

can be calculated with the formula

$$S_n = \frac{a(1 - q^n)}{1 - q}.$$

Here $n \in \mathbb{Z}^+$.

4. Let $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ be the direction vector of line L going through the point $A = (2, -2, 4)$, and let P be any point on the line. The vector equation of the line is

$$\overrightarrow{OP} = \overrightarrow{OA} + t\mathbf{v},$$

where $t \in \mathbb{R}$.

5. In parametric form,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} + t \begin{pmatrix} -1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 - t \\ -1 + 2t \\ -2 - 4t \end{pmatrix}, \text{ where } t \in \mathbb{R}.$$

6. The system of equations

$$\begin{aligned}x - y + z &= 2 \\ -x + 4y + 2z &= -2 \\ 2x + y - z &= 14\end{aligned}$$

corresponds to the augmented matrix

$$\begin{pmatrix} 1 & -1 & 1 & 2 \\ -1 & 4 & 2 & -2 \\ 2 & 1 & -1 & 14 \end{pmatrix}.$$